

Introduction to Networking Fundamentals

Module 1.0

Brendan Shea, PhD

Rochester Community and Technical College
Intro to Networking



Module Overview

- This module introduces the fundamental concepts you need to understand computer networks.
- We will explore different types of networks and how they are physically and logically organized.
- The **OSI Model** provides a framework for understanding how data travels across networks.
- **SOHO networks** demonstrate how these concepts apply to small office and home environments.
- A systematic **troubleshooting methodology** helps diagnose and resolve network problems efficiently.

Topics Covered

- 1 Networking Overview
- 2 OSI Model Concepts
- 3 SOHO Networks
- 4 Troubleshooting

Learning Outcomes

After completing this module, you will be able to:

- Define fundamental **networking concepts** and terminology used by IT professionals.
- Identify different **network types** including LAN, WAN, PAN, MAN, and CAN.
- Describe common **network topologies** and explain their advantages and disadvantages.
- Explain the seven layers of the **OSI model** and the function of each layer.
- Describe the components and functions of a **SOHO router**.
- Apply a systematic **troubleshooting methodology** to diagnose network problems.

Key Terms Preview

Network Basics

- **Node** – Any device on a network
- **Host** – A device with an IP address
- **Topology** – Network layout/design
- **Protocol** – Rules for communication

Network Types

- **LAN** – Local Area Network
- **WAN** – Wide Area Network
- **SOHO** – Small Office/Home Office

OSI & Data

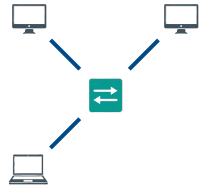
- **OSI Model** – 7-layer reference model
- **Encapsulation** – Wrapping data with headers
- **PDU** – Protocol Data Unit

Number Systems

- **Binary** – Base-2 (0 and 1)
- **Hexadecimal** – Base-16 (0–F)
- **MAC Address** – Hardware address

What is a Network?

- A **computer network** is two or more devices connected together to share resources and communicate.
- Networks allow users to share files, printers, internet connections, and applications across multiple devices.
- Every network requires three basic components: **nodes** (devices), **links** (connections), and **protocols** (rules).
- Networks can be as simple as two laptops connected via Bluetooth or as complex as the global internet.



Simple Network

Why Networks Matter

Networks enable collaboration, centralized data storage, resource sharing, and communication—the foundation of modern business and daily life.

Networking Concepts

- A **node** is any device connected to a network, including computers, printers, and smartphones.
- A **host** is a specific type of node that has an IP address and can send or receive data.
- **Clients** are devices that request resources or services from other computers on the network.
- **Servers** are computers that provide resources or services to clients, such as web pages, files, or email.
- In a **peer-to-peer** network, devices act as both clients and servers, sharing resources directly with each other.

Client-Server Example

When you visit a website, your computer (client) sends a request to a web server, which responds by sending the web page back to your browser.

Network Types

- A **LAN (Local Area Network)** connects devices in a small geographic area like a home, office, or school building.
- A **WAN (Wide Area Network)** connects LANs across large geographic distances, often using leased telecommunication lines.
- A **PAN (Personal Area Network)** connects devices within a person's immediate workspace, typically within 10 meters.
- A **MAN (Metropolitan Area Network)** spans a city or campus, larger than a LAN but smaller than a WAN.
- A **CAN (Campus Area Network)** connects multiple LANs within a limited geographic area like a university or corporate campus.

The Internet

The **internet** is the largest WAN in existence—a global network of interconnected networks using standardized protocols.

Network Types Comparison

Type	Range	Example	Typical Speed
PAN	<10 meters	Bluetooth headset	1–3 Mbps
LAN	Building/floor	Office network	100 Mbps–10 Gbps
CAN	Campus	University network	1–10 Gbps
MAN	City	City library system	100 Mbps–1 Gbps
WAN	Global	Corporate branches	1 Mbps–1 Gbps

Size Relationship

$\text{PAN} \subset \text{LAN} \subset \text{CAN} \subset \text{MAN} \subset \text{WAN}$

Speed vs. Distance

Generally, smaller networks offer faster speeds due to shorter cable runs and fewer devices.

What is Network Topology?

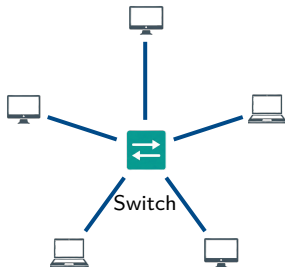
- **Network topology** describes the arrangement of nodes and connections in a network.
- **Physical topology** refers to the actual layout of cables, devices, and other hardware components.
- **Logical topology** describes how data flows through the network, regardless of physical layout.
- Understanding topology helps network administrators design efficient networks and troubleshoot problems.
- The choice of topology affects cost, performance, scalability, and fault tolerance of a network.

Physical vs. Logical Example

A network might be physically wired in a star pattern (all cables to a central switch), but logically function as a bus where all devices receive all transmissions.

Star Topology

- In a **star topology**, all nodes connect to a central device such as a switch or hub.
- This is the most common topology used in modern LANs due to its simplicity and reliability.
- If one node or cable fails, only that device is affected—other nodes continue to function normally.
- The central device is a **single point of failure**; if it fails, the entire network goes down.

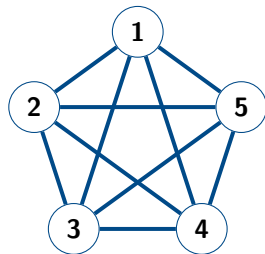


Advantages

Easy to install, manage, and troubleshoot; simple to add new devices.

Mesh Topology

- In a **mesh topology**, nodes are interconnected with multiple redundant paths between them.
- A **full mesh** connects every node directly to every other node, providing maximum redundancy.
- A **partial mesh** connects only some nodes with multiple paths, balancing cost and redundancy.
- Mesh topologies are highly fault-tolerant because data can take alternate routes if a link fails.



Full Mesh (10 links)

Formula: Full Mesh Links

Links needed = $\frac{n(n-1)}{2}$ where n = number of nodes.

Example: 5 nodes require $\frac{5 \times 4}{2} = 10$ links.

Legacy Topologies

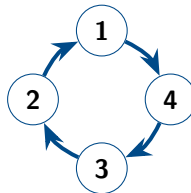
Bus Topology

- All nodes connect to a single central cable called the **backbone** or **bus**.
- Signals travel along the bus and are received by all nodes.
- A break anywhere in the cable disables the entire network.



Ring Topology

- Each node connects to exactly two other nodes, forming a circular data path.
- Data travels in one direction (or both in a **dual ring**).
- A single node failure can break the entire ring.



Case Study: Mystery Inc. Opens a Detective Agency

► Case Study: Mystery Inc. Network Setup

The Mystery Inc. gang is opening a new detective agency headquarters and needs to set up a computer network for five workstations.

Shaggy suggests: “Like, let’s just connect every computer to every other computer, man. That way if Scooby chews through one cable, we’re still connected!”

Velma responds: “Jinkies! That’s overkill. Let’s connect everything to one central switch instead—it’s much more practical.”

Review Questions

- 1 What network topology is Shaggy describing?
- 2 What network topology is Velma describing?
- 3 Which topology would be more practical for a small office, and why?
- 4 How many cables would Shaggy’s design require for 5 computers?

Case Study Solution: Mystery Inc. Network Setup

✓ Solution: Mystery Inc. Network Setup

- ① Shaggy is describing a **full mesh topology**, where every device connects directly to every other device.
- ② Velma is describing a **star topology**, where all devices connect to a central switch.
- ③ **Star topology** is more practical for a small office because:
 - Requires fewer cables (5 vs. 10)
 - Easier to install and manage
 - Simpler to add new devices later
 - More cost-effective for small networks
- ④ Shaggy's full mesh would require $\frac{5 \times 4}{2} = 10$ cables.

Key Takeaway

While mesh topologies offer excellent redundancy, star topologies are the standard choice for most LANs due to their balance of simplicity, cost, and reliability.

The OSI Model

- The **OSI (Open Systems Interconnection) Model** is a conceptual framework that standardizes network communication into seven layers.
- Developed by the International Organization for Standardization (ISO) in 1984 to promote interoperability between different vendors.
- Each layer performs specific functions and communicates with the layers directly above and below it.
- The OSI model is a **reference model**—it describes concepts, not a specific implementation or protocol.
- Understanding the OSI model is essential for troubleshooting because it helps isolate where problems occur.

The Seven Layers (Bottom to Top)

1. Physical
2. Data Link
3. Network
4. Transport
5. Session
6. Presentation
7. Application

Why the OSI Model Matters

- The OSI model provides a **common vocabulary** that allows network professionals to communicate clearly about network functions.
- It enables **vendor interoperability** by defining standard interfaces between layers that different manufacturers can implement.
- The layered approach supports **modular design**, allowing changes to one layer without affecting others.
- Network administrators use the OSI model as a **troubleshooting framework** to systematically isolate problems.

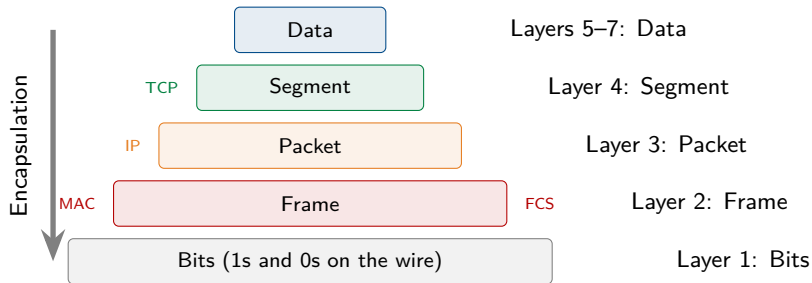
Troubleshooting Example

“The cable is fine (Layer 1), and the switch sees the MAC address (Layer 2), but there’s no IP address (Layer 3).”

This pinpoints the problem to Layer 3—likely a DHCP issue.

Data Encapsulation and Decapsulation

- **Encapsulation** is the process of wrapping data with protocol headers (and sometimes trailers) as it moves down the OSI layers.
- Each layer adds its own control information to the data received from the layer above.
- **Decapsulation** is the reverse process—removing headers as data moves up the layers at the receiving device.
- The data unit at each layer has a specific name called a **Protocol Data Unit (PDU)**.



Layer 1: Physical

- The **Physical layer** is responsible for transmitting raw **bits** (1s and 0s) over a physical medium.
- It defines electrical, mechanical, and procedural specifications for cables, connectors, and signaling.
- Physical layer specifications include voltage levels, timing, data rates, and maximum transmission distances.
- This layer has no understanding of data meaning—it simply moves bits from one device to another.

Layer 1 Devices

- Cables (copper, fiber)
- Connectors (RJ-45)
- **Hubs**
- **Repeaters**
- Network adapters
- Wireless radios

PDU: Bits

At Layer 1, data is represented as individual bits transmitted as electrical signals, light pulses, or radio

WAVES

Layer 2: Data Link

- The **Data Link layer** packages bits into **frames** and provides node-to-node communication on the same network.
- It uses **MAC (Media Access Control) addresses** to identify devices—unique 48-bit hardware addresses burned into network cards.
- This layer handles error detection using a **Frame Check Sequence (FCS)** to verify data integrity.
- The Data Link layer is divided into two sublayers: **LLC** (Logical Link Control) and **MAC**.

MAC Address Format

00:1A:2B:3C:4D:5E — Six pairs of hexadecimal digits (48 bits total).

Layer 2 Devices

- **Switches**
- **Bridges**
- Wireless access points
- Network interface cards

PDU: Frame

Frames contain source and destination MAC addresses.

Layer 3: Network

- The **Network layer** handles **logical addressing** and **routing** of data between different networks.
- It uses **IP (Internet Protocol) addresses** to identify devices across interconnected networks.
- **Routing** is the process of selecting the best path for data to travel from source to destination.
- This layer enables communication beyond the local network—it's what makes the internet possible.

IP Address Examples

IPv4: 192.168.1.100 (32 bits)

IPv6: 2001:0db8:85a3::8a2e:0370:7334 (128 bits)

Layer 3 Devices

- **Routers**
- Layer 3 switches
- Firewalls (often)

PDU: Packet

Packets contain source and destination IP addresses.

Layer 4: Transport

- The **Transport layer** provides end-to-end communication between applications running on different hosts.
- It uses **port numbers** (0–65535) to identify specific applications or services on a device.
- **TCP (Transmission Control Protocol)** provides reliable, connection-oriented delivery with error correction and flow control.
- **UDP (User Datagram Protocol)** provides fast, connectionless delivery without guaranteed delivery or ordering.

TCP vs. UDP

TCP: Web browsing, email, file transfer—when accuracy matters.

UDP: Video streaming, VoIP, online gaming—when speed matters.

Common Ports

HTTP: 80

HTTPS: 443

SSH: 22

DNS: 53

PDU: Segment (TCP) / Datagram (UDP)

Transport layer PDUs contain source and destination port numbers.

Upper Layers: Session, Presentation, Application

Layer 5: Session

- Manages **sessions** (dialogs) between applications
- Establishes, maintains, and terminates connections
- Handles authentication and reconnection

Examples: NetBIOS, RPC, SQL sessions

Layer 6: Presentation

- Translates data formats between applications
- Handles **encryption** and decryption
- Manages **compression** and character encoding

Examples: SSL/TLS, JPEG, ASCII, MPEG

Layer 7: Application

- Closest layer to the end user
- Provides network services to applications
- Interfaces directly with user software

Examples: HTTP, FTP, SMTP, DNS, DHCP

Important Note

In practice, the upper layers (5, 6, 7) are often combined in the TCP/IP model as a single “Application” layer, since many protocols span multiple OSI layers.

OSI Model Summary

Layer	Name	PDU	Key Devices	Protocols/Examples
7	Application	Data	Hosts, firewalls	HTTP, FTP, SMTP, DNS
6	Presentation	Data	Hosts	SSL/TLS, JPEG, MPEG
5	Session	Data	Hosts	NetBIOS, RPC
4	Transport	Segment	Hosts, firewalls	TCP, UDP
3	Network	Packet	Routers, L3 switches	IP, ICMP, ARP
2	Data Link	Frame	Switches, bridges	Ethernet, Wi-Fi (802.11)
1	Physical	Bits	Hubs, cables, NICs	Ethernet (physical), DSL

Lower Layers (1–4)

Handle data transport and delivery—the “plumbing” of the network.

Upper Layers (5–7)

Handle data representation and application services—closer to the user.

OSI Memory Tricks

Layer 1 → Layer 7

Please **D**o **N**ot **T**hrow **S**ausage **P**izza **A**way

1. **P**hysical
2. **D**ata Link
3. **N**etwork
4. **T**ransport
5. **S**ession
6. **P**resentation
7. **A**pplication

Layer 7 → Layer 1

All **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing

7. **A**pplication
6. **P**resentation
5. **S**ession
4. **T**ransport
3. **N**etwork
2. **D**ata Link
1. **P**hysical

Quick PDU Mnemonic

Don't **S**ome **P**eople **F**ear **B**irthdays? — **D**ata, **S**egment, **P**acket, **F**rame, **B**its (Layers 7→1)

Case Study: The Haunted Network

► Case Study: The Haunted Network

Daphne reports that she cannot access any websites from her computer at Mystery Inc. headquarters.

Fred runs some diagnostic tests and finds:

- Her computer has an IP address (192.168.1.105)
- She can successfully ping the router (192.168.1.1)
- She can successfully ping 8.8.8.8 (Google's DNS server)
- She **cannot** access `www.mysteryinc.com` or any other website

Review Questions

- 1 Which OSI layers appear to be working correctly based on Fred's tests?
- 2 At which OSI layer is the problem most likely occurring?
- 3 What might be causing this issue?

Case Study Solution: The Haunted Network

✓ Solution: The Haunted Network

① Layers 1–3 are working correctly:

- Layer 1 (Physical): Cable connected, signals transmitting
- Layer 2 (Data Link): Frames reaching the router
- Layer 3 (Network): IP assigned, can route to internet (8.8.8.8)

② The problem is most likely at **Layer 7 (Application)** or involves **DNS resolution**.

③ Possible causes include:

- DNS server is down or misconfigured
- Browser misconfiguration or corruption
- Firewall blocking HTTP/HTTPS (ports 80/443)
- Malware redirecting web traffic

Key Takeaway

The OSI model helps isolate problems layer by layer—if Layer 3 works (ping by IP), check Layers 4–7 next.

What is a SOHO Network?

- **SOHO** stands for **Small Office/Home Office** and refers to networks with fewer than 10 users.
- SOHO networks are designed to be simple, affordable, and easy to manage without dedicated IT staff.
- A typical SOHO network connects computers, printers, smartphones, and tablets to share internet access and resources.
- Most SOHO networks use a single multi-function device that combines several networking functions.

Common SOHO Devices

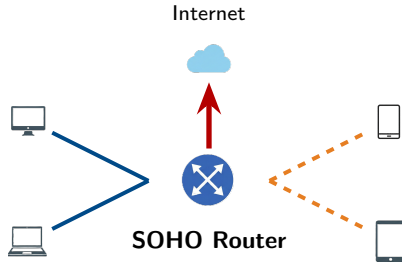
- Desktop computers
- Laptops
- Smartphones
- Tablets
- Printers
- Smart TVs
- IoT devices

SOHO Examples

Home networks, small retail shops, home-based businesses, small professional offices (law, accounting, medical).

SOHO Router Overview

- A **SOHO router** is an all-in-one device that combines multiple networking functions into a single unit.
- Despite being called a “router,” these devices typically include a router, switch, wireless access point, and sometimes a modem.
- SOHO routers provide a simple way to connect a small network to the internet with minimal configuration.
- Most SOHO routers include basic security features such as a firewall and support for wireless encryption.



SOHO Router: Physical Layer Functions

- At Layer 1, the SOHO router provides physical connections for both wired and wireless devices.
- Most SOHO routers include 4–8 **RJ-45 Ethernet ports** for connecting wired devices like computers and printers.
- A separate **WAN port** connects to the ISP's modem or directly to the internet service.
- The built-in **wireless radio** transmits and receives data using radio frequencies (typically 2.4 GHz and 5 GHz).
- LED indicators provide visual feedback about power, connectivity, and network activity.

Physical Components

- Ethernet ports (LAN)
- WAN/Internet port
- Wi-Fi antennas
- Power connector
- Reset button
- Status LEDs

Common Speeds

Fast Ethernet: 100 Mbps
Gigabit: 1000 Mbps
Wi-Fi 6: up to 9.6 Gbps

SOHO Router: Data Link Layer Functions

- At Layer 2, the SOHO router functions as a **switch** for its LAN ports, forwarding frames based on MAC addresses.
- The router maintains a **MAC address table** that maps each device's MAC address to a specific port.
- For wireless connections, the router handles **802.11 frame formatting** and manages wireless client associations.
- **MAC filtering** allows administrators to permit or deny network access based on device MAC addresses.
- The router can also implement **VLANs** to segment the network into separate broadcast domains.

Layer 2 Features

- Switching between LAN ports
- MAC address learning
- Wireless encryption (WPA3, WPA2)
- MAC address filtering
- Guest network isolation

Security Tip

MAC filtering alone is not secure—MAC addresses can be spoofed. Always use encryption!

SOHO Router: Network Layer Functions

- At Layer 3, the SOHO router performs its primary function: **routing** packets between the LAN and the internet.
- **NAT (Network Address Translation)** allows multiple devices to share a single public IP address from the ISP.
- The built-in **DHCP server** automatically assigns IP addresses, subnet masks, and gateway information to LAN devices.
- The router acts as the **default gateway** for all devices on the local network.
- Most SOHO routers use **private IP addresses** (e.g., 192.168.1.x) for the internal network.

Layer 3 Features

- NAT/PAT translation
- DHCP server
- Default gateway
- Static routing
- DNS relay/proxy

Private IP Ranges

10.0.0.0/8
172.16.0.0/12
192.168.0.0/16

SOHO Router: Transport, Application, and Security

Transport Layer (4)

- **Port forwarding** directs incoming traffic on specific ports to designated internal devices.
- **Port triggering** dynamically opens ports when outbound traffic is detected.
- **UPnP** (Universal Plug and Play) allows applications to automatically configure port mappings.

Application Layer (7)

- Web-based management interface
- DNS relay services
- QoS (Quality of Service) prioritization

Security Features

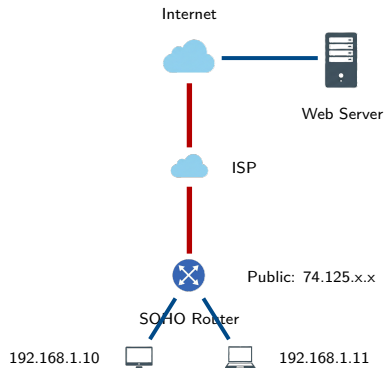
- **Stateful firewall** tracks connection states and blocks unsolicited inbound traffic.
- **WPA3/WPA2 encryption** protects wireless communications.
- **Content filtering** can block access to specific websites or categories.
- **Parental controls** restrict access by time or content type.
- **VPN passthrough** or VPN server capabilities.

The Internet

- The **internet** is a global network of interconnected networks that use standardized protocols (TCP/IP) to communicate.
- Your SOHO router connects your local network to your **ISP (Internet Service Provider)**, which connects to larger networks.
- Data travels through multiple networks and routers to reach its destination, a process called **routing**.
- Your ISP assigns your router a **public IP address** that identifies your network on the internet.

Public vs. Private IP

Private IPs work only on your LAN; public IPs are routable on the internet. NAT translates between them.



Why Binary and Hexadecimal Matter

- Computers process everything as **binary** (1s and 0s), including all network addresses and data.
- **IP addresses** are actually 32-bit binary numbers, though we write them in decimal for convenience.
- **MAC addresses** are 48-bit binary numbers written in hexadecimal notation.
- Understanding binary helps you work with **subnet masks**, calculate network ranges, and troubleshoot addressing issues.

The Same Address in Three Formats

Binary:	11000000.10101000.00000001.00001010
Decimal:	192.168.1.10
Hexadecimal:	C0.A8.01.0A

Coming Up

Let's learn how to convert between these number systems.

Binary Number System

- The **binary** (base-2) system uses only two digits: 0 and 1.
- Each digit is called a **bit**; eight bits form a **byte** (0–255 in decimal).
- Place values double from right to left: 1, 2, 4, 8, 16, 32, 64, 128.
- To convert to decimal: multiply each bit by its place value, then add.

Example: Converting 11010110 to Decimal

128	64	32	16	8	4	2	1
1	1	0	1	0	1	1	0
128	64	–	16	–	4	2	–
128 + 64 + 16 + 4 + 2 = 214							

Memorize the Powers of 2

$2^0 = 1$, $2^1 = 2$, $2^2 = 4$, $2^3 = 8$, $2^4 = 16$, $2^5 = 32$, $2^6 = 64$, $2^7 = 128$

Hexadecimal Number System

- **Hexadecimal** (base-16) uses digits 0–9 and letters A–F (where A=10 through F=15).
- Each hex digit represents exactly 4 bits, making it a compact way to write binary.
- MAC addresses use hex:
00:1A:2B:3C:4D:5E

Binary to Hex

Convert 11010110:

Split: 1101 | 0110

Convert: D | 6

Result: **0xD6**

4-Bit Reference

Dec	Bin	Hex
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Case Study: Scooby Snacks Café Wi-Fi

► Case Study: Scooby Snacks Café Wi-Fi

Shaggy has opened “Scooby Snacks Café” and set up a SOHO router to provide free Wi-Fi for customers.

A customer connects to the Wi-Fi and receives the IP address 192.168.1.50. They can successfully print to the café’s wireless printer at 192.168.1.25, but they cannot access any websites.

Velma checks the router’s status page and sees the WAN port shows “Disconnected.”

Review Questions

- 1 Is 192.168.1.50 a public or private IP address?
- 2 Which SOHO router function successfully assigned this IP address?
- 3 Why can the customer reach the printer but not websites?
- 4 What should Shaggy check to fix the internet connectivity?

Case Study Solution: Scooby Snacks Café Wi-Fi

✓ Solution: Scooby Snacks Café Wi-Fi

- ❶ 192.168.1.50 is a **private IP address** (from the 192.168.0.0/16 private range).
- ❷ The **DHCP server** function is working correctly—it assigned a valid IP to the customer's device.
- ❸ The customer can reach the printer because both devices are on the **same local network** (Layer 2 switching works). Websites require internet access through the **WAN connection**, which is down.
- ❹ Shaggy should check:
 - Is the cable from the WAN port to the modem connected?
 - Is the ISP modem powered on and working?
 - Did Shaggy pay the internet bill?

Key Takeaway

A SOHO router combines many functions. When troubleshooting, identify which functions work (DHCP, switching) and which don't (WAN/internet) to isolate the problem.

Network Troubleshooting Methodology

- CompTIA defines a **7-step troubleshooting methodology** that provides a systematic approach to diagnosing and resolving network problems.
- Following a structured process saves time and ensures problems are fully resolved, not just temporarily fixed.
- This methodology applies to all types of technical problems, not just networking issues.
- Documentation at each step creates a knowledge base for solving similar problems in the future.

The 7 Steps

- 1 Identify the problem
- 2 Establish a theory
- 3 Test the theory
- 4 Establish a plan
- 5 Implement the solution
- 6 Verify functionality
- 7 Document findings

Step 1: Identify the Problem

- The first step is to gather information and clearly define what is wrong before attempting any fixes.
- **Question users** to understand the symptoms—ask what they were doing, when it started, and what error messages appeared.
- **Identify symptoms** by observing the problem firsthand whenever possible.
- **Determine the scope**—is this affecting one user, one department, or the entire network?
- **Check for recent changes**—new software, hardware, or configuration changes often cause problems.

Key Questions to Ask

- What exactly is happening?
- When did it start?
- Has anything changed recently?
- Who is affected?
- Can you reproduce it?
- What have you already tried?

Step 2: Establish a Theory of Probable Cause

- Based on the symptoms, develop one or more **theories** about what might be causing the problem.
- Start with the most likely or simplest explanation—this principle is called **Occam's Razor**.
- Use the **OSI model** as a framework: start at Layer 1 (physical) and work up, or start at Layer 7 and work down.
- Consider multiple possibilities and rank them by likelihood before testing.
- If you're stuck, research the symptoms online or consult documentation.

Common Causes

- Loose or damaged cables
- Incorrect IP settings
- DHCP issues
- DNS problems
- Firewall blocking traffic
- Hardware failure

Occam's Razor

“The simplest explanation is usually correct.” Check the obvious things first—is it plugged in? Is it turned on?

Step 3: Test the Theory to Determine the Cause

- Once you have a theory, **test it** to confirm or eliminate it as the cause.
- Use diagnostic tools and commands to gather evidence (ping, traceroute, ipconfig, etc.).
- If your theory is **confirmed**, proceed to Step 4 (establish a plan of action).
- If your theory is **not confirmed**, return to Step 2 and establish a new theory.
- Sometimes testing requires making small changes—always note what you change so you can undo it.

Theory Confirmed

Theory: Cable is bad.

Test: Swap cable.

Result: Connection works!

→ Proceed to Step 4.

Theory Not Confirmed

Theory: Cable is bad.

Test: Swap cable.

Result: Still doesn't work.

→ Return to Step 2.

Steps 4 & 5: Establish a Plan and Implement the Solution

Step 4: Establish a Plan of Action

- Create a clear plan to resolve the problem based on your confirmed theory.
- Consider the **impact** on users—can you fix it now, or schedule downtime?
- Identify resources needed (parts, software, expertise).
- Get necessary **approvals** before making changes.
- Have a **rollback plan** in case the fix doesn't work.

Step 5: Implement the Solution

- Execute your plan, making changes carefully and methodically.
- Change **one thing at a time** so you know what fixed the problem.
- If the fix is beyond your skill level, **escalate** to senior staff or vendors.
- Keep notes on exactly what you changed.

Steps 6 & 7: Verify Functionality and Document Findings

Step 6: Verify Full Functionality

- Confirm the original problem is resolved—don't assume!
- Test thoroughly: have the **user verify** the fix works for them.
- Check that your fix didn't **break anything else**.
- Implement **preventive measures** to stop the problem from recurring.

Step 7: Document Findings

- Record the **problem symptoms** and how they were reported.
- Document the **root cause** you identified.
- List all **actions taken** to resolve the issue.
- Note the **outcome** and any follow-up needed.

Why Document?

Good documentation helps you solve similar problems faster, trains other staff, provides evidence for management, and builds an organizational knowledge base.

Troubleshooting Methodology Summary

Step	Name	Key Actions
1	Identify the problem	Question users, identify symptoms, determine scope
2	Establish a theory	Consider probable causes, use OSI model, Occam's Razor
3	Test the theory	Confirm or eliminate; if wrong, return to Step 2
4	Establish a plan	Plan the fix, consider impact, get approvals
5	Implement solution	Execute plan, escalate if needed
6	Verify functionality	Confirm fix works, check for side effects
7	Document findings	Record problem, cause, actions, and outcome

Memory Tip

I Eat Tacos Every Instant Very Deliciously

(Identify, Establish, Test, Establish, Implement, Verify, Document)

Case Study: The Mystery Machine Won't Connect

► Case Study: The Mystery Machine Won't Connect

The gang is on a stakeout in the Mystery Machine. Velma's laptop suddenly cannot connect to Fred's mobile hotspot, but Daphne's tablet connects without any problems.

Velma has already tried rebooting her laptop, but it still won't connect. She sees the hotspot name in her Wi-Fi list, but gets an "Unable to connect" error when she tries to join.

Review Questions (Apply the 7-Step Methodology)

- ❶ **Step 1:** Based on the information given, what is the scope of this problem?
- ❷ **Step 2:** Using Occam's Razor, list two simple theories for the probable cause.
- ❸ **Step 3:** What test could Velma perform to confirm or eliminate one of your theories?

Case Study Solution: The Mystery Machine Won't Connect

✓ Solution: The Mystery Machine Won't Connect

- ① **Step 1 (Scope):** The problem affects only Velma's laptop. Daphne's tablet works fine, and the hotspot itself is functioning—this points to an issue with Velma's device specifically.
- ② **Step 2 (Theories):** Simple explanations include:
 - Velma's saved password for the hotspot is incorrect (Fred may have changed it)
 - Velma's Wi-Fi adapter is malfunctioning
 - The hotspot has a device limit and Velma's laptop is blocked
- ③ **Step 3 (Test):** Velma could “forget” the saved network and re-enter the password. If that works, the theory is confirmed. If not, try connecting to a different Wi-Fi network to test if her adapter works at all.

Key Takeaway

Determining the **scope** (one device vs. many) quickly narrows down where the problem lies. Start with simple tests before assuming hardware failure.

Module 1.0 Summary

Networking Fundamentals

- Networks connect devices to share resources using nodes, links, and protocols.
- Types: PAN, LAN, CAN, MAN, WAN (smallest to largest).
- Topologies: Star (common), mesh (redundant), bus/ring (legacy).

OSI Model

- Seven layers: Physical, Data Link, Network, Transport, Session, Presentation, Application.

SOHO Networks

- SOHO routers combine router, switch, wireless AP, DHCP, and firewall.
- NAT allows private IPs to share one public IP.
- Binary (base-2) and hex (base-16) represent network addresses.

Troubleshooting (7 Steps)

- Identify → Theory → Test → Plan → Implement → Verify → Document.